Principles of Data Communications and Networks Modeling (CS4865/6865)

Assignment #1: Theoretical Channel Capacity

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Question 1: 802.11b Wireless Channel

Given Data

- Transmitter power: $P_t = 0.1$ W
- Channel bandwidth: $B = 22$ MHz $= 22 \times 10^6$ Hz
- Carrier frequency: $f = 2.4$ GHz $= 2.4 \times 10^9$ Hz
- Thermal noise temperature: $T = 400$ K
- Boltzmann Constant: $k = 1.38 \times 10^{-23}\ J/K$
- Speed of light: $c = 3 \times 10^8\ m/s$
- Distance: $d = 10$ m to 100 m
- Pi: $\pi = 3.14$

Calculations

The thermal noise is given by $N = kTB$, where k is the Boltzmann constant, T is the temperature in Kelvin, and B is the bandwidth. Using given values $k = 1.38 \times 10^{-23}$ J/K, $T = 400$ K, and $B = 22 \times 10^6$ Hz, the thermal noise is:

$$N = (1.38 \times 10^{-23}) \times 400 \times (22 \times 10^6) \approx 1.21 \times 10^{-13} \text{ W}$$

The free space loss for isotropic antenna is given by $L = \frac{P_t}{P_r}$. From Friis transmission equation [2], we can also denote $L = (\frac{4\pi f d}{c})^2$, where f is the carrier frequency, d is the distance between transmitter and receiver, c is the speed of light.

For distance 10m:

Free space loss is denoted by L,

$$L = (\frac{4\pi f d}{c})^2 = [\frac{4 \times 3.14 \times (2.4 \times 10^9) \times 10}{3 \times 10^8}]^2 \approx 1.01 \times 10^6$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01 \times 10^6} \approx 9.9 \times 10^{-8}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{9.9\times10^{-8}}{1.21\times10^{-13}})\approx 4.32\times10^8$$

For distance 20m:

Free space loss is denoted by L,

$$L = (\frac{4\pi fd}{c})^2 = [\frac{4\times3.14\times(2.4\times10^9)\times20}{3\times10^8}]^2\approx 4.04\times10^6$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01\times10^6}\approx 2.48\times10^{-8}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{2.48\times10^{-8}}{1.21\times10^{-13}})\approx 3.88\times10^8$$

For distance 30m:

Free space loss is denoted by L,

$$L = (\frac{4\pi fd}{c})^2 = [\frac{4\times3.14\times(2.4\times10^9)\times30}{3\times10^8}]^2\approx 9.09\times10^6$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01\times10^6}\approx 1.10\times10^{-8}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{1.10\times10^{-8}}{1.21\times10^{-13}})\approx 3.62\times10^8$$

For distance 40m:

Free space loss is denoted by L,

$$L = (\frac{4\pi fd}{c})^2 = [\frac{4\times3.14\times(2.4\times10^9)\times40}{3\times10^8}]^2\approx 1.62\times10^7$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01\times10^6}\approx 6.19\times10^{-9}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{6.19\times10^{-9}}{1.21\times10^{-13}})\approx 3.44\times10^8$$

For distance 50m:

Free space loss is denoted by L,

$$L = (\frac{4\pi f d}{c})^2 = [\frac{4 \times 3.14 \times (2.4 \times 10^9) \times 50}{3 \times 10^8}]^2 \approx 2.52 \times 10^7$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01 \times 10^6} \approx 3.96 \times 10^{-9}$$

Shannon capacity formula:

$$C = B \log_2 \left(1 + \frac{S}{N}\right) = (22 \times 10^6) \times \log_2(1 + \frac{3.96 \times 10^{-9}}{1.21 \times 10^{-13}}) \approx 3.3 \times 10^8$$

For distance 60m:

Free space loss is denoted by L,

$$L = (\frac{4\pi f d}{c})^2 = [\frac{4 \times 3.14 \times (2.4 \times 10^9) \times 60}{3 \times 10^8}]^2 \approx 3.63 \times 10^7$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01 \times 10^6} \approx 2.75 \times 10^{-9}$$

Shannon capacity formula:

$$C = B \log_2 \left(1 + \frac{S}{N}\right) = (22 \times 10^6) \times \log_2(1 + \frac{2.75 \times 10^{-9}}{1.21 \times 10^{-13}}) \approx 3.18 \times 10^8$$

For distance 70m:

Free space loss is denoted by L,

$$L = (\frac{4\pi f d}{c})^2 = [\frac{4 \times 3.14 \times (2.4 \times 10^9) \times 70}{3 \times 10^8}]^2 \approx 4.95 \times 10^7$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01 \times 10^6} \approx 2.02 \times 10^{-9}$$

Shannon capacity formula:

$$C = B \log_2 \left(1 + \frac{S}{N}\right) = (22 \times 10^6) \times \log_2(1 + \frac{2.02 \times 10^{-9}}{1.21 \times 10^{-13}}) \approx 3.09 \times 10^8$$

For distance 80m:

Free space loss is denoted by L,

$$L = (\frac{4\pi f d}{c})^2 = [\frac{4 \times 3.14 \times (2.4 \times 10^9) \times 80}{3 \times 10^8}]^2 \approx 6.46 \times 10^7$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01 \times 10^6} \approx 1.55 \times 10^{-9}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{1.55\times10^{-9}}{1.21\times10^{-13}}) \approx 3.00\times10^8$$

For distance 90m:

Free space loss is denoted by L,

$$L = (\frac{4\pi fd}{c})^2 = [\frac{4\times3.14\times(2.4\times10^9)\times90}{3\times10^8}]^2 \approx 8.18\times10^7$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01\times10^6} \approx 1.22\times10^{-9}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{1.22\times10^{-9}}{1.21\times10^{-13}}) \approx 2.93\times10^8$$

For distance 100m:

Free space loss is denoted by L,

$$L = (\frac{4\pi fd}{c})^2 = [\frac{4\times3.14\times(2.4\times10^9)\times100}{3\times10^8}]^2 \approx 1.01\times10^8$$

Received signal strength is denoted by S,

$$S = P_r = \frac{P_t}{L} = \frac{0.1}{1.01\times10^6} \approx 9.9\times10^{-10}$$

Shannon capacity formula:

$$C = B\log_2\left(1+\frac{S}{N}\right) = (22\times10^6)\times\log_2(1+\frac{9.9\times10^{-10}}{1.21\times10^{-13}}) \approx 2.86\times10^8$$

1.1 Data table of numerical values of distance and data rate

Below is the table with numerical values of distance and data rate which were calculated at 10m interval.

Serial No.	Distance (m)	Data rate (Mbps)
1	10	432
2	20	388
3	30	362
4	40	344
5	50	330
6	60	318
7	70	309
8	80	300
9	90	293
10	100	286

Table 1: Numerical table of distance and data rate

1.2 Plot of Maximum Data Rate (Theoretical) vs Distance

After plotting the values that was calculated earlier, the following graph can be observed.

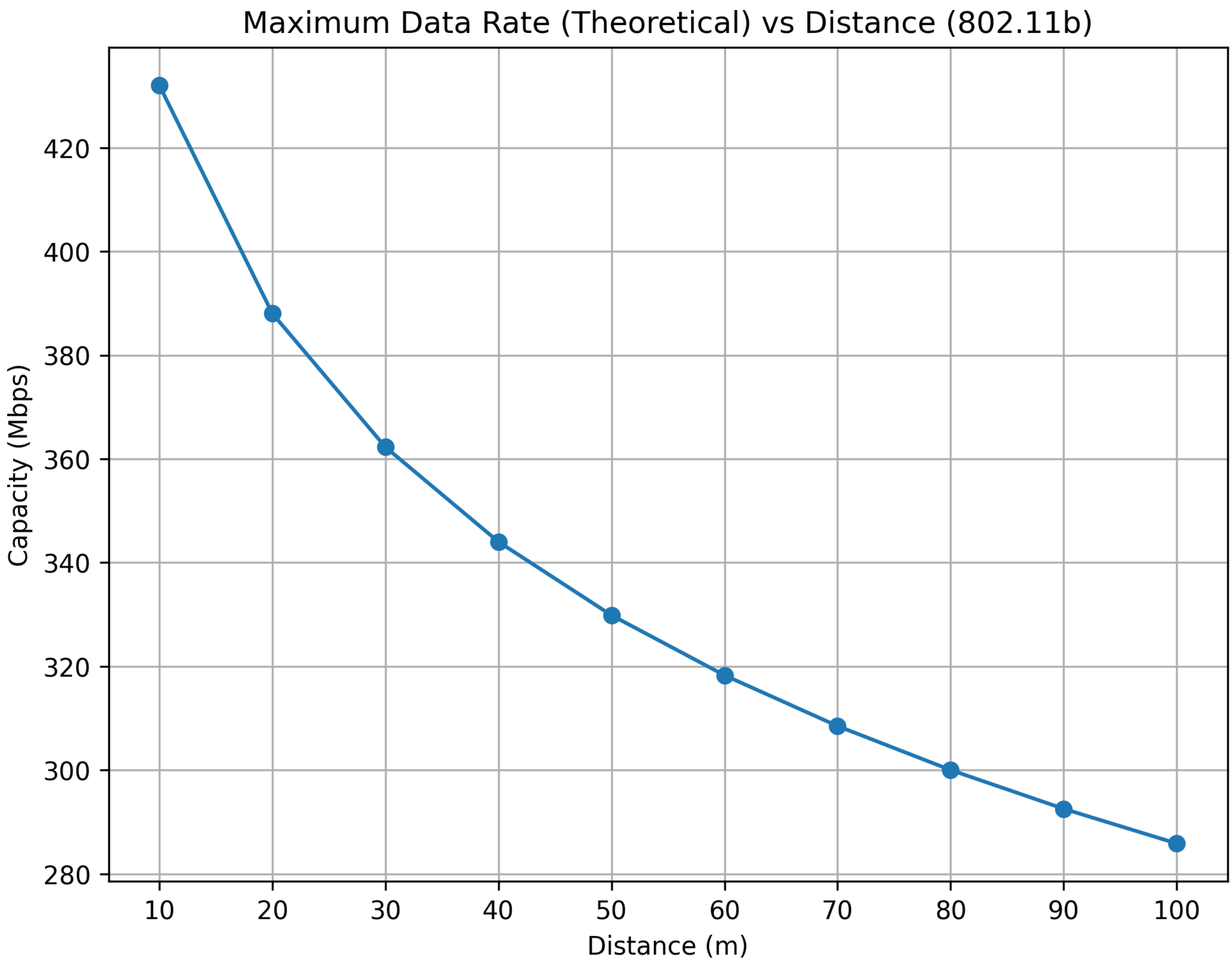

Figure 1: Maximum data rate (theoretical) for distances from 10m to 100 m

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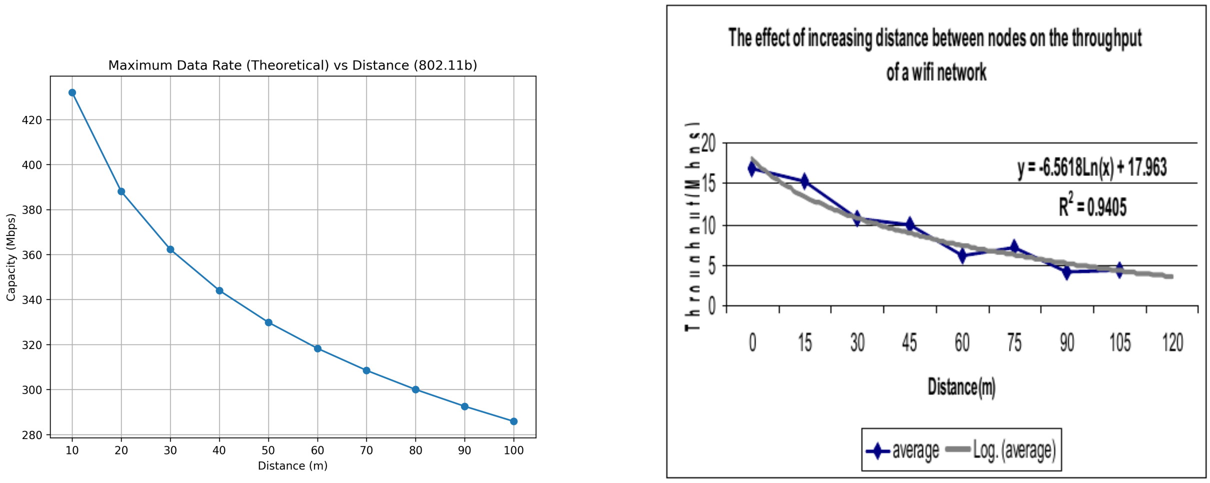
Question 2: Comparison with the given research paper (Experimental Performance Evaluation of Wireless 802.11b Networks)

2.1 Comparison of measured values

The measured values from the mentioned research paper and the calculated theoretical ones differ quite a lot. For reference, two graphs are shown below -



(a) Maximum data rate (theoretical)

(b) Measured data rate from paper

Figure 2: Value comparison of data rates between theoretical and practical

2.2 Discussion of theoretical throughput and measured values in paper

Overall, the theoretical and measured values are not same. There are both similarities and dissimilarities in between theoretical throughput and measured values.

2.2.1 Similarities

- Both theoretical and measured data rate graphs show downward trend as distance increases. This is expected with wireless behavior where the signal strength decreases with increased distance.

2.2.2 Dissimilarities

- **Parameter Differences:** The parameters which were used to calculate the data rate of theoretical and measured one differs, such as - temperature, distance, network traffic etc. For theoretical, 10m interval is considered whereas in measured one it was 15m. Also, in case of theoretical data rate, interference was not considered as there was no real network traffic.
- **Experimental Devices:** In case of theoretical, a single device was considered. But in the measured one, there were three separate devices. In practical one, it introduced interference and performance was varied. Also a software *Iperf* was used to measure the performance in case of the measured one. Devices used in the mentioned paper were not identical and thus their specifications were different which are absent in the theoretical one.
- **Protocol and Modulation Overheads:** In case of theoretical, the data rate calculation does not consider protocol overheads, such as modulation or CSMA/CA delays [1]. But in measured one, these factors introduced additional latency and thus reduced the overall throughput which also lead to difference between theoretical and measured values.

The above points are my justification for why the theoretical calculated and measured values from the research paper differs.

Question 3: Mars Rover Communication

Given Data

- Average Earth to Mars distance, $d = (\frac{100{,}000{,}000+400{,}000{,}000}{2})$ km $= 2.5 \times 10^{11}$ m
- Transmitter Power (Curiosity), $P_t = 15$ W
- LGA gain (Curiosity), $G_t = 5$ dBi
- Transmitter Power (DSN), $P_t = 20$ kW
- Downlink carrier frequency (X-band), $f = 8.4$ GHz
- DSN receive antenna gains (on-axis, X-band), $G_{r,34} = 68.3$ dBi, $G_r, 70 = 74.18$ dBi
- Frequency (DTE - Downlink), $f_{dte} = 8.4$ GHz
- Frequency (DFE - Uplink), $f_{dfe} = 7.2$ GHz
- System noise temperature (receiver), $T_{sys,34} = 27$ K, $T_{sys,70} = 17$ K
- Bandwidth, $B = 500$ kHz
- Boltzmann Constant, $k = 1.38 \times 10^{-23}\ J/K$
- Speed of light, $c = 3 \times 10^8\ m/s$
- Transmitter power, frequency, receiver noise, distance to Earth

3.1 Calculations

3.1.1 Space Loss:

For DTE (8.4 GHz): The space loss is denoted by L_s,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{8.4 \times 10^9} = 0.0357 \text{ m}$$

$$L_s = (\frac{\lambda}{4\pi d})^2 = (\frac{0.0357}{4\pi \times 2.5 \times 10^{11}}) = 1.29 \times 10^{-28}$$

For DFE (7.2 GHz): The space loss is denoted by L_s,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.2 \times 10^9} = 0.0417 \text{ m}$$

$$L_s = (\frac{\lambda}{4\pi d})^2 = (\frac{0.0417}{4\pi \times 2.5 \times 10^{11}}) = 1.76 \times 10^{-28}$$

3.1.2 Signal to Noise ratio:

For DTE: The signal to noise ratio is denoted by SNR_{dte}.

$$SNR = \frac{P_t G_t G_r L_s L_o}{k T_{sys} B}$$

After converting gains to linear scale we get, $G_t = 10^{5/10} = 3.16$ and $G_r = 10^{68/10} = 6.31 \times 10^6$

$$SNR_{dte} = \frac{15 \times 3.16 \times (6.31 \times 10^6) \times (1.29 \times 10^{-28}) \times 0.5}{(1.38 \times 10^{-23} \times 25 \times 500000)} = 1.12 \times 10^{-4}$$

For DFE: The signal to noise ratio is denoted by SNR_{dfe}.

$$SNR = \frac{P_t G_t G_r L_s L_o}{kT_{sys}B}$$

After converting gains to linear scale we get, $G_t = 10^{5/10} = 3.16$ and $G_r = 10^{68/10} = 6.31 \times 10^6$

$$SNR_{dfe} = \frac{20000 \times (6.31 \times 10^6) \times 3.16 \times (1.76 \times 10^{-28}) \times 0.5}{(1.38 \times 10^{-23} \times 25 \times 500000)} = 0.203$$

3.2 Theoretical Maximum Data Rate Calculation

3.2.1 DTE Data Rate:

$$C_{dte} = 50000 \times \log_2(1 + 1.12 \times 10^{-4}) = 80.78bps$$

3.2.2 DFE Data Rate:

$$C_{dfe} = 50000 \times \log_2(1 + 0.203) = 13331.83bps$$

References

References

[1] WiFi standard 802.11b: `https://en.wikipedia.org/wiki/IEEE_802.11b-1999`

[2] Friis transmission equation: `https://en.wikipedia.org/wiki/Friis_transmission_equation`

[3] A.B. Fadhah et al., *Experimental performance evaluation of Wireless 802.11b networks*, 2008 ICADIWT

[4] NASA, Communicating with Curiosity: `https://science.nasa.gov/resource/communicating-with-curiosity-2/`

[5] S. Gordon, Communications with Mars Curiosity: `https://sandilands.info/sgordon/communications-with-mars-curiosity`